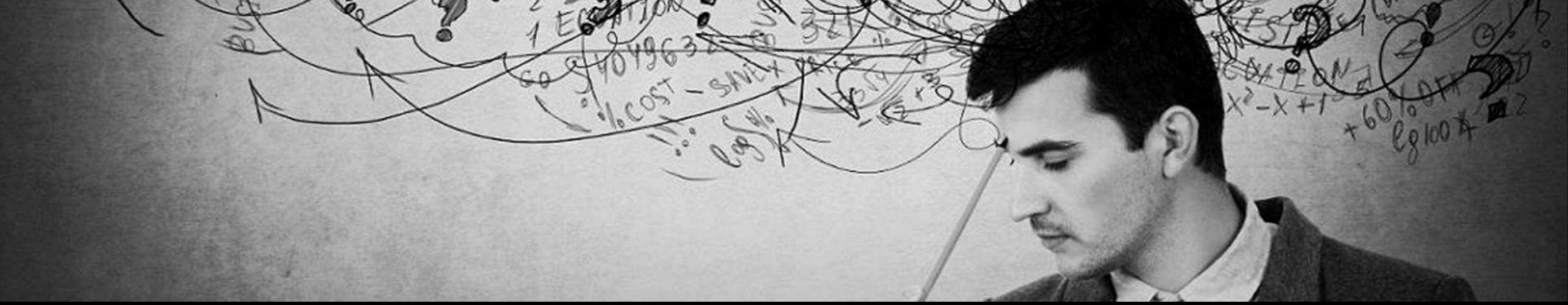
DILEMMA

2025-01 midterm

Group 1



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Introduction :That is Paradox/Dilemma?

Haon

The dilemma is not just to make difficult choices, but to make meaningful choice. In many situations we are asked to decide between two equally valuable outcomes. There may not be a clear "right" or "wrong," just a competitive form of line. That's why it makes the dilemma so important and difficult. They challenge us to think beyond our instincts or rules. They ask us: What is the most important thing? To whom? And why? When exploring paradoxes and dilemmas, we explore ourselves, our priorities and principles.

Heize dilemma &Security dilemma

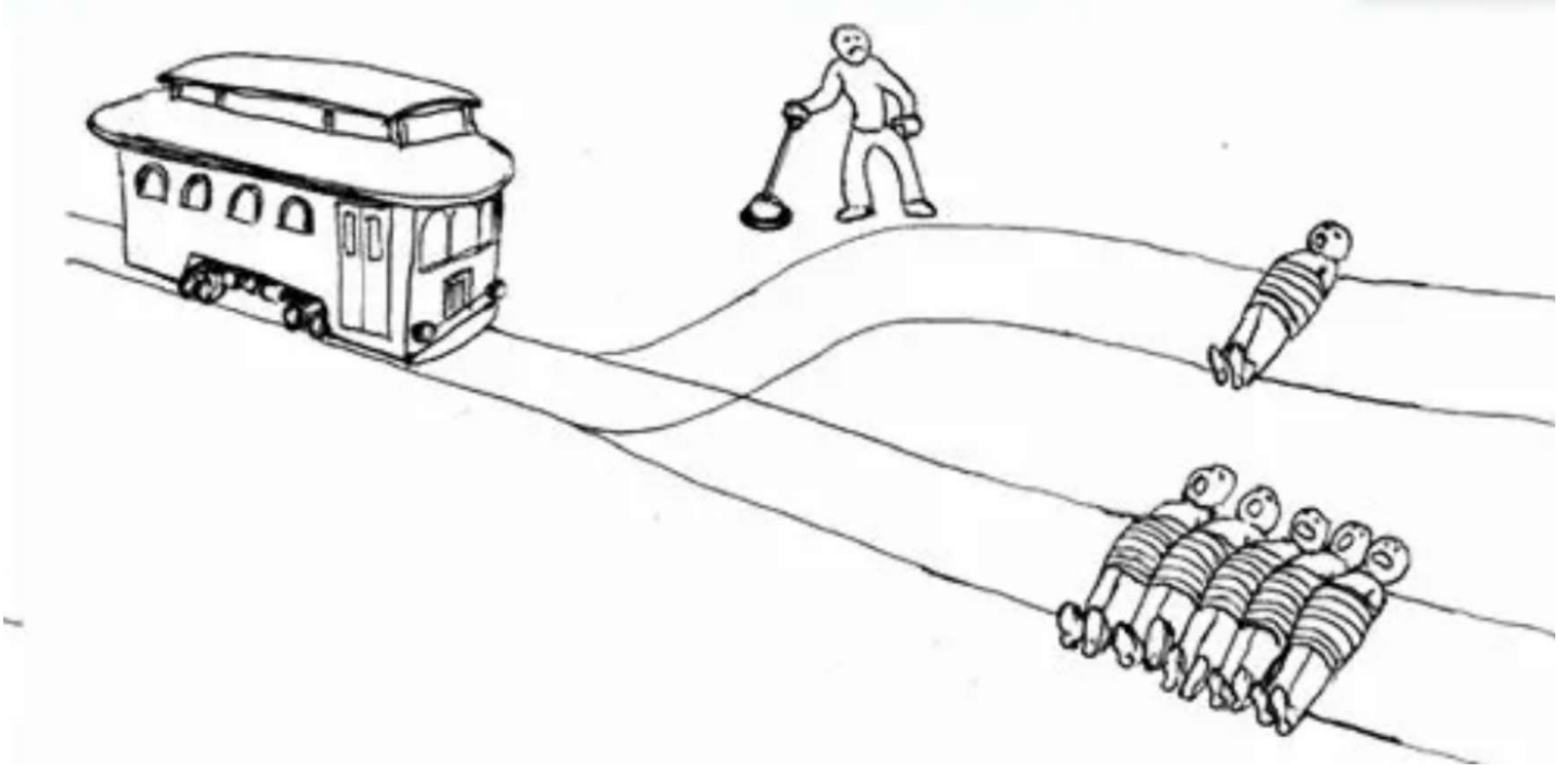
Haon

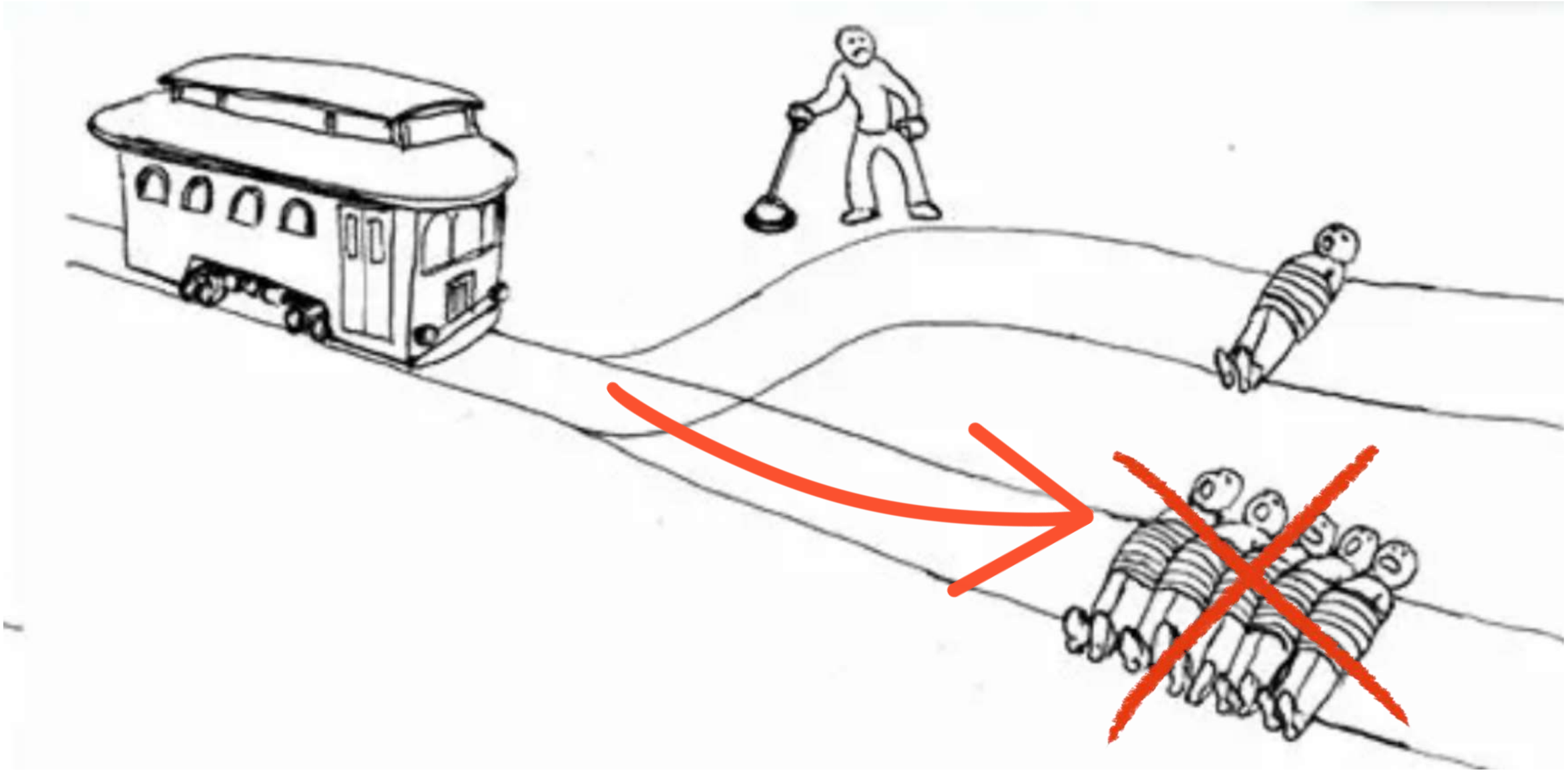
The Heinz dilemma asks if breaking the law is justified in saving lives, and the security dilemma shows how defensive actions can be considered a threat. Both raise deeper questions about whether good intentions can lead to detrimental results. In Heinz's case, life-saving moral values conflict with the rule of law. Attempts by states to protect themselves in global politics can cause fear and tension in others. This dilemma reminds us that ethical decisions are often shaped not only by actions but also by how we perceive such actions. Understanding both can show how complex real morality can be.

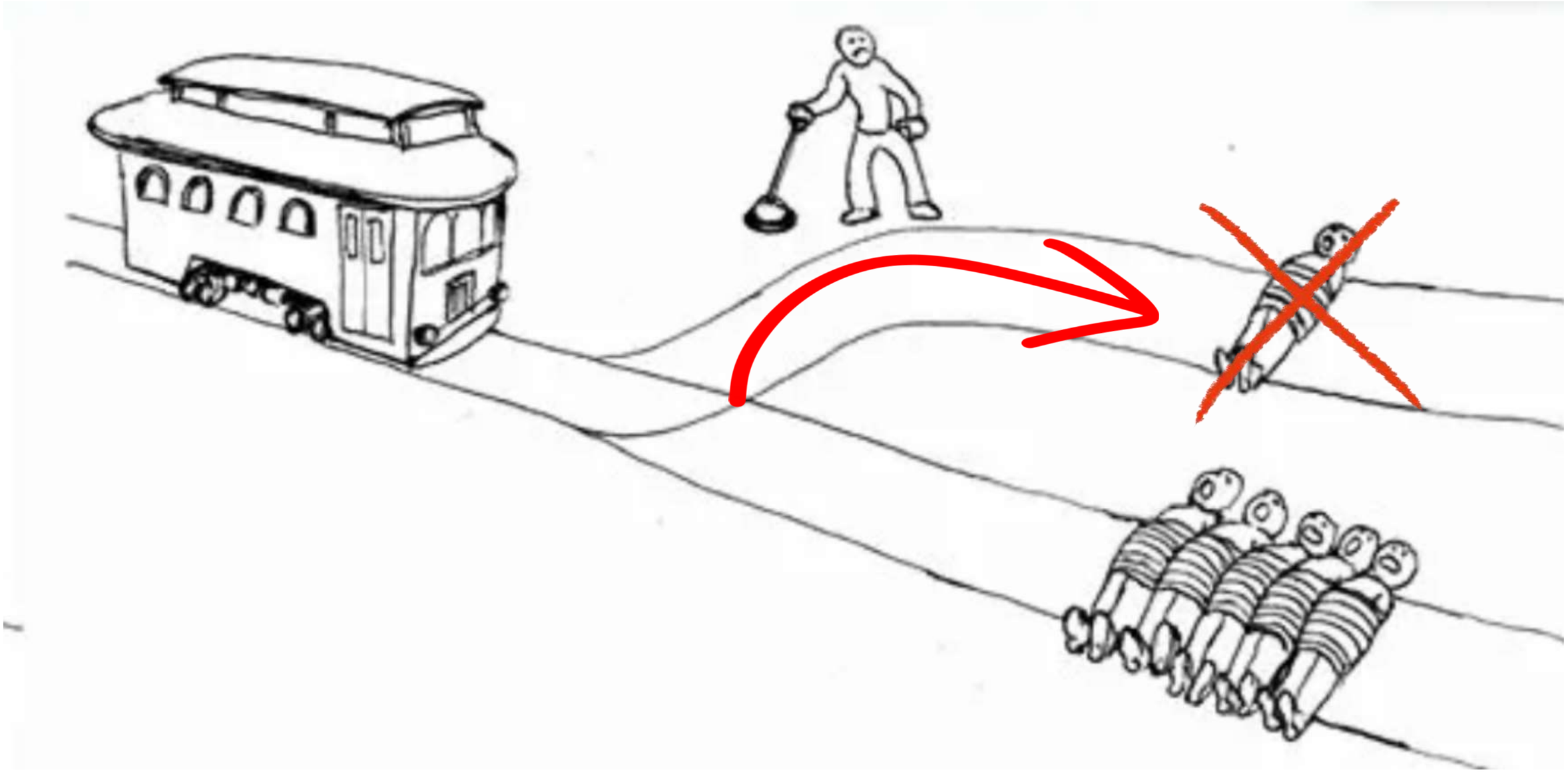
Trolley Dilemma

Dahyun

The trolley dilemma creates a moral conflict between action and inaction. The fleeing trolley threatens to kill five people on the front track. Pulling the lever to switch to another track kills one person instead. Should we act to minimize the damage, or should we refuse to intervene and avoid direct responsibility? This scenario raises deep questions about ethics, intentions, and outcomes. It is not just about numbers but how to define right and wrong. This thought experiment prompts us to look at the principles of moral choice.







Cicero

UTILITARIANISM

-Pull the lever.

-the greatest happiness of the greatest number

-Focus on the results.

THE THEORY OF OBLIGATION

-Do not pull the lever.

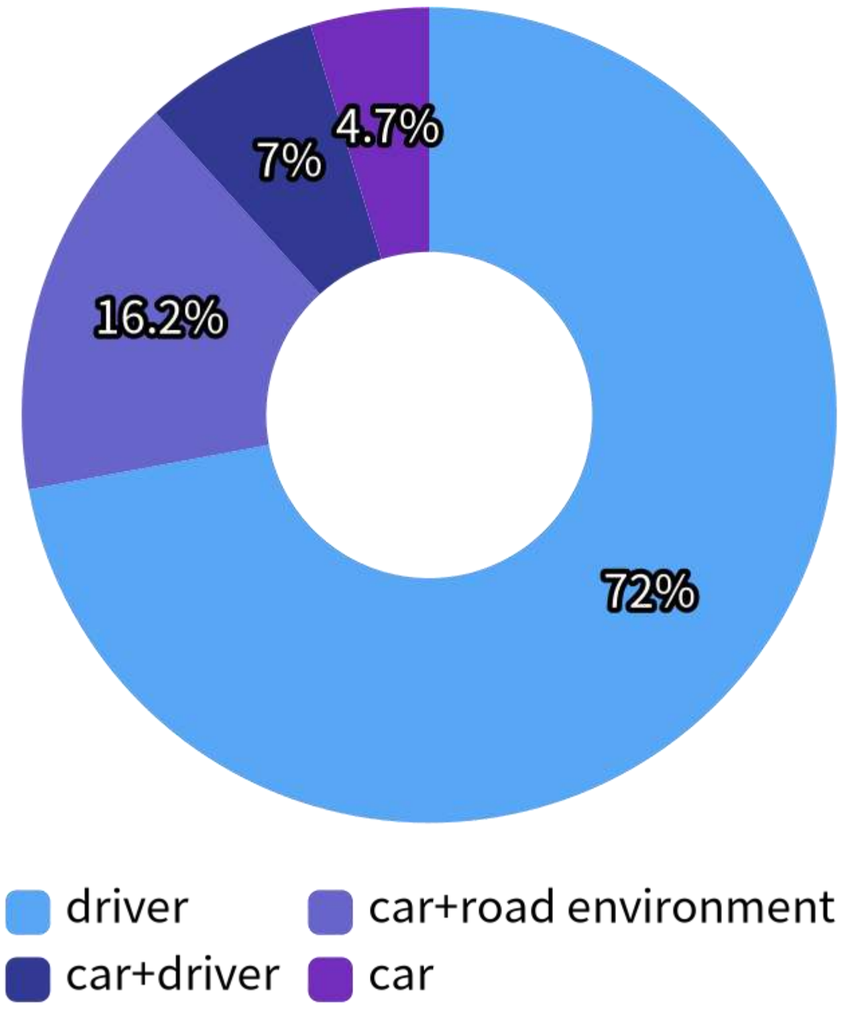
-Treat humans as an end, not as a means.

-Focus on process and intent.

Autonomous Vehicle Dilemma

eunkyo

Autonomous vehicle cars pose a modern version of the classic trolley dilemma. If collisions are inevitable, should cars protect passengers or minimize overall damage? Should cars be sacrificed to save many people, even if they harm pedestrians? This dilemma forces us to program ethical decisions into machines. This obscures the line between human judgment and algorithmic logic. Ethics becomes code, and every line comes with consequences. In shaping AI choices, we face our own values and fears.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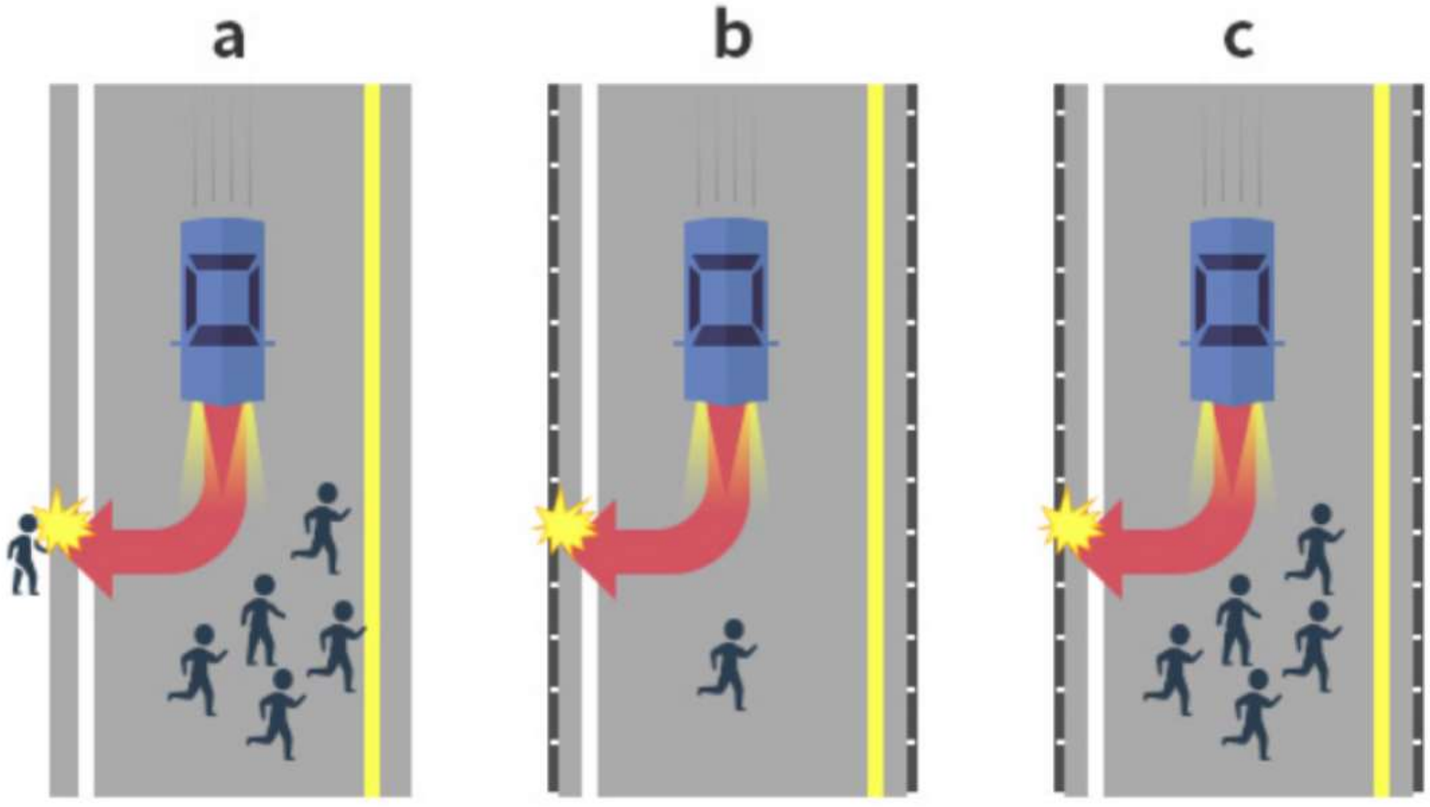
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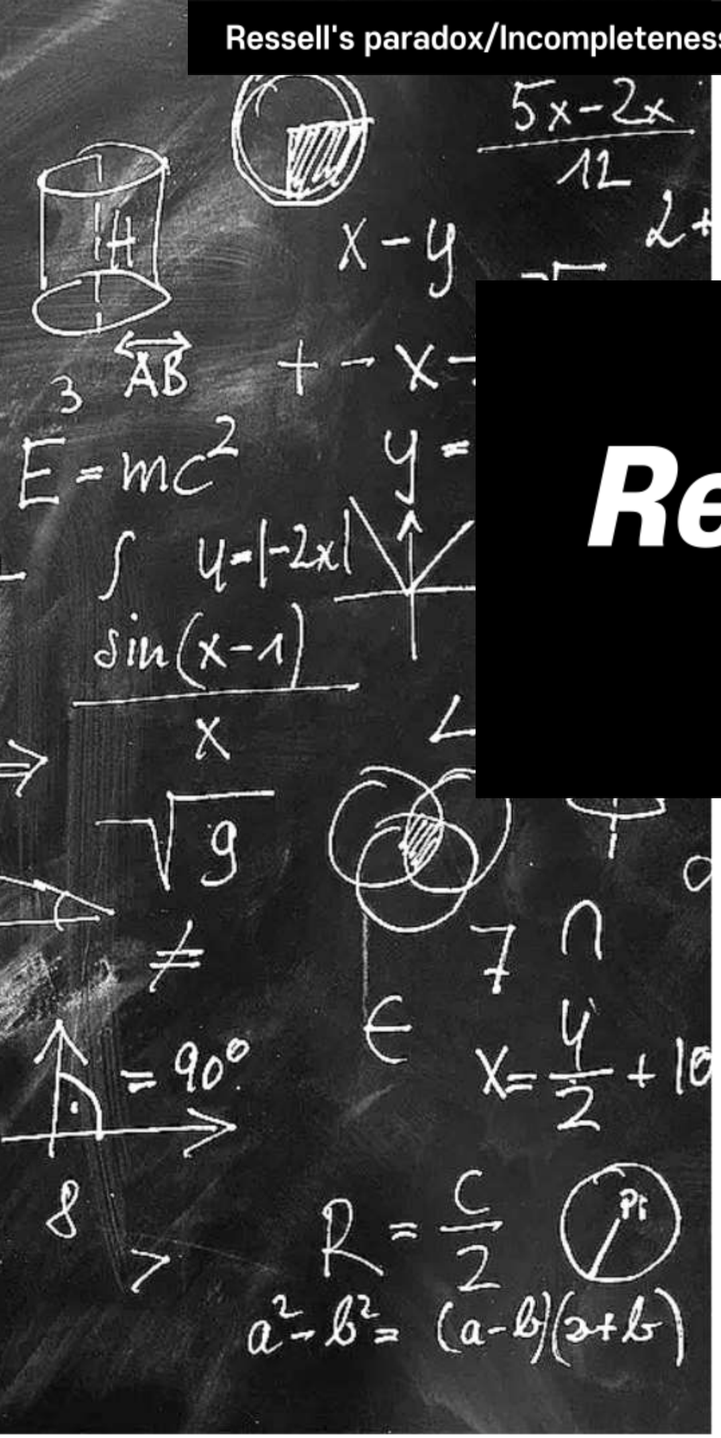


Ressell's Paradox

woosang

Russell's Paradox reveals a contradiction in naive set theory by asking whether the set of all sets that do not contain themselves includes itself. If it does, it shouldn't; if it doesn't, it should — leading to a logical loop with no consistent answer. This paradox showed that even basic assumptions in mathematics could lead to serious problems when self-reference is involved. It forced philosophers and mathematicians to rethink how sets, truth, and definitions should work. More than a technical issue, it reflects the limits of formal reasoning. Russell's insight became a turning point in the foundations of logic and mathematics.

Barber Paradox

rule:

The barber shaves

all and only those men in the town

who do not shave themselves.

Barber Paradox as a Dilemma

Q. Does the barber shave himself?

-If the barber shaves himself,

he must not shave himself.

-If the barber does not shave himself,

he must shave himself.

No matter what you choose, you run into trouble

⟶ a logical dilemma

When Shaving Becomes a Set Theory Problem

This funny puzzle is actually

similar to a serious problem in math

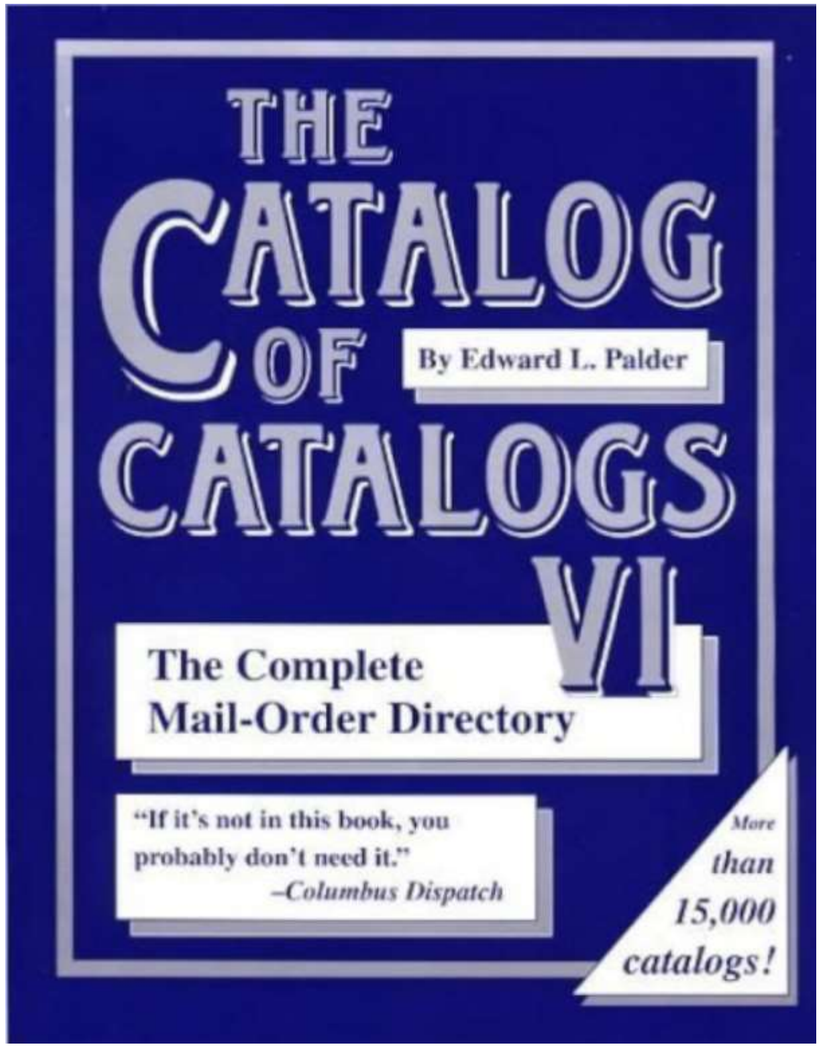
Russell's Paradox arises

when we consider the set:

$$R = \{x \mid x \notin x\},$$

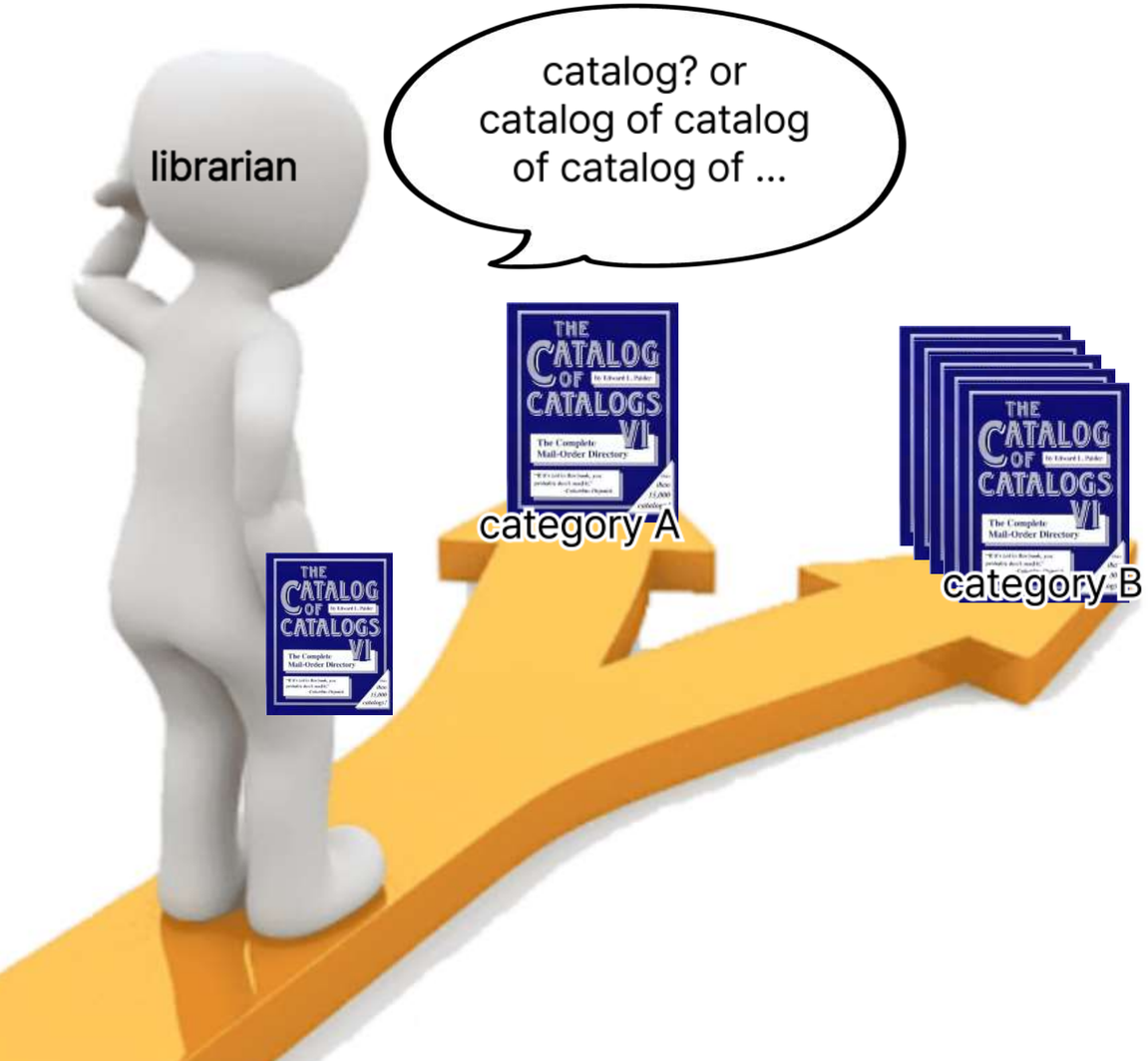
the set of all sets that do not contain themselves.

(imagine a catalog)



When Shaving Becomes a Set Theory Problem

Q. Does the barber shave himself?



Now ask: Does R contain itself?

-If $R \in R$, then by definition, $R \notin R$.

-If $R \notin R$, then by definition, $R \in R$.

→ a logical dilemma again

When we encounter dilemmas

We face two big challenges:

1. No clear resolution: Just like with the Barber or Russell's paradox, sometimes logical systems break down when we try to apply simple rules to complex problems.

2. Complexity in decision-making: In both ethics and mathematics, dilemmas force us to think deeply about the foundations and the limits of the systems we use.

Why These Paradoxes/Dilemmas Matter

-These paradoxes help us understand the limits of our reasoning and systems.

-By studying these dilemmas, we can improve the frameworks we use to make decisions — whether in ethics or in mathematics.

Why These Paradoxes/Dilemmas Matter

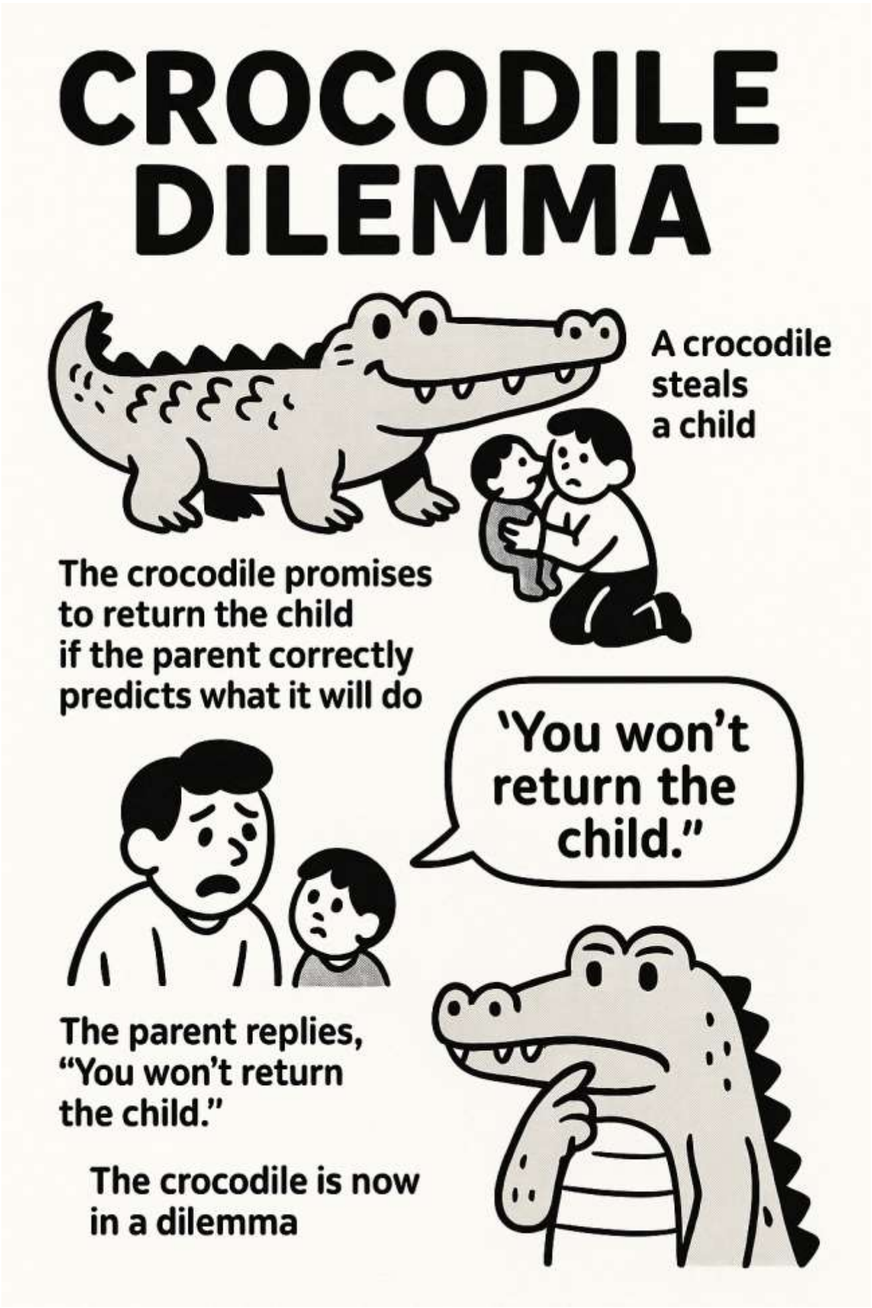
-We see how small choices can lead to big problems

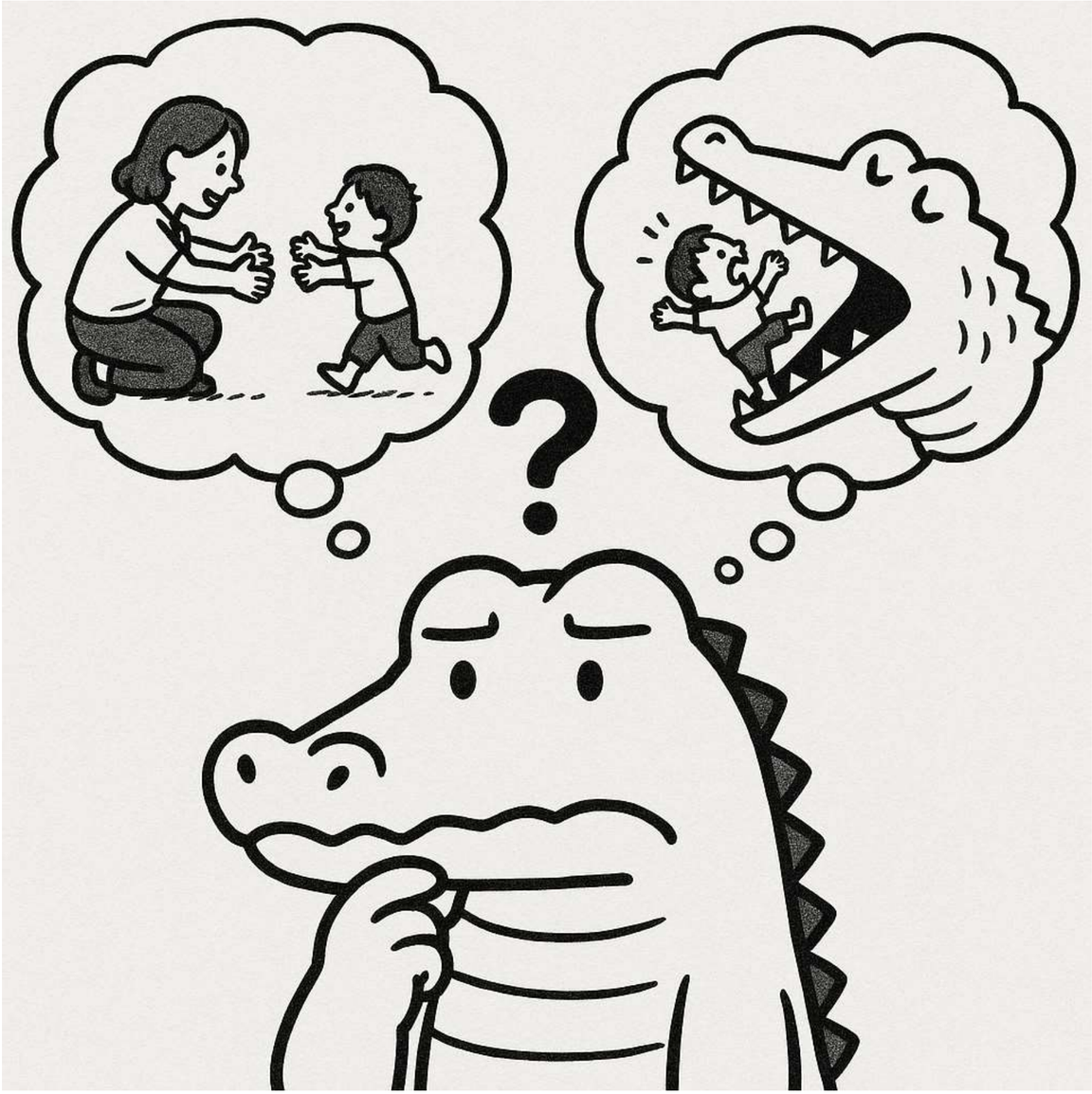
-Understanding these paradoxes helps us improve how we think about complex problems and make better decisions.

Crocodile's dilemma

Alex

The crocodile's dilemma explains the logical paradox that a crocodile steals a child and promises to return it only when parents guess exactly what the crocodile will do. If parents say "I'm not returning my child," and the crocodile plans to return the child, it shouldn't return the child because the guess is wrong. However, if the crocodile now decides not to return the child, the crocodile should return the child because the parent's guess is correct. This creates a self-reference loop without a consistent solution, showing that language and prediction can break logic when tied together


CROCODILE DILEMMA
A crocodile steals a child
The crocodile promises to return the child if the parent correctly predicts what it will do
'You won't return the child."
The parent replies, "You won't return the child."
The crocodile is now in a dilemma



Conclusion

Alex

As we have seen, paradoxes and dilemmas allow us to revisit the principles, rules, and judgment standards we are familiar with. In particular, moral dilemmas or logical paradoxes make us fundamentally question what criteria we choose and what we think is 'right'. "By what standards do we think and judge?" With these questions, we learn to think a little bit more maturely. Thank you for listening.

THANK YOU FOR LISTENING

Haon Dahyun Eunkyo Woosang Alex